

Maternal Health Risk Stratification

Short Clinical Analysis Summary

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Objective

This summary compares Low, Mid, and High maternal risk groups and evaluates a simple multinomial Logistic Regression model. Particular attention is given to errors involving High Risk observations rather than relying on overall accuracy alone.

Data Quality

- Raw project file: 1,014 observations, 7 variables, no missing values.
- 562 exact duplicate rows were identified.
- Two records with HeartRate = 7 bpm were excluded because the values were physiologically implausible.
- Exact repeated rows were removed for the modeling analysis, leaving 451 unique observations. Because patient identifiers are unavailable, this decision is treated as a methodological assumption.
- Thirty five identical predictor profiles were associated with more than one RiskLevel label. These observations were retained because the disagreement may reflect label ambiguity or clinical information that is not recorded in the dataset.

Main Findings

Finding	Result
Strongest High Risk separation	Blood Sugar, SMD = 1.61
Next strongest variables	SystolicBP = 0.70; DiastolicBP = 0.61
Baseline model	Multinomial Logistic Regression + StandardScaler pipeline
Split	80/20 stratified; random_state = 42
Accuracy	0.670
High Risk recall	0.565 (13/23 detected)
Mid Risk recall	0.048 (1/21 detected)
Most common error	Mid Risk to Low Risk, 17 cases
Missed High Risk error	High Risk to Low Risk, 10 cases
Largest High Risk coefficient	BS = +1.176

Interpretation

Blood Sugar shows the strongest descriptive separation and the largest positive contribution to the High Risk model score. High Risk observations without markedly elevated Blood Sugar are more difficult for the model to

detect. Overall accuracy therefore gives an incomplete picture of performance. The model performs especially poorly for Mid Risk and misses 43.5% of the High Risk observations in this test split.

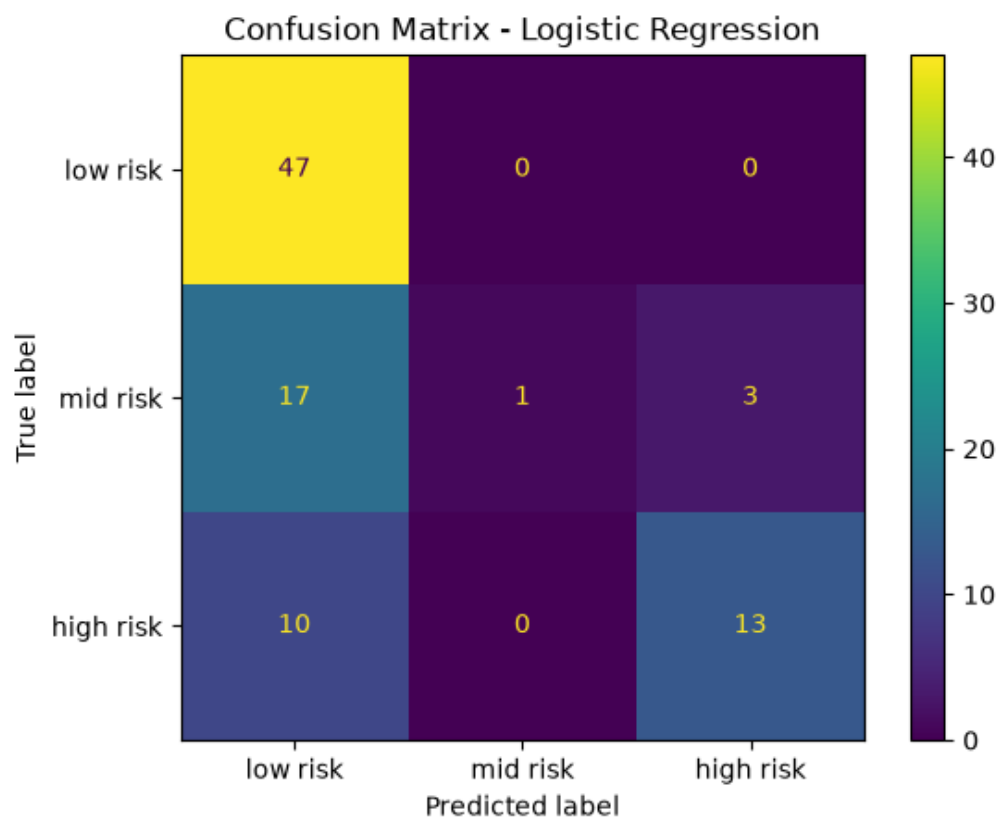
Limitations

- Small cleaned sample and only six predictors.
- Duplicate handling cannot be verified at the patient level.
- Conflicting labels exist for identical predictor profiles.
- The results are based on one training and test split, with no external validation.
- The reported associations and coefficients should not be interpreted as causal clinical effects.

Conclusion

The analysis provides an educational baseline for maternal health risk classification. Its main value is the examination of class specific recall, false negative errors, preprocessing assumptions, and the limitations of the available data. The model is not intended for diagnosis or clinical decision making.

Key Visual



Confusion matrix for the Logistic Regression baseline. Most errors involve Mid Risk observations classified as Low Risk and High Risk observations classified as Low Risk.